CSE 216 Formal Methods
Fall 2026 Daniel Fremont

HW 4

Due October 27 at 3:00 pm

(2 questions, 110 points total)

1. (40 pts.) EUF

Use the congruence closure algorithm to decide which of the following EUF conjunctions is satisfiable. Show the steps of the algorithm.

(a) $f(x) = x \wedge f(g(x)) \neq g(x)$

Answer: the initial partition has 4 terms

$$\{\{f(x)\},\{x\},\{f(g(x))\},\{g(x)\}\} \tag{1}$$

there is only one new equivalence class

$$\{\{f(x),x\},\{f(g(x))\},\{g(x)\}\} \tag{2}$$

none of the non equalities cause a contradiction with the equivalence class created by the equalities thus the EUF conjunction is sat.

(b) $f(x,y) = f(y,x) \wedge g(y) \neq g(f(y,g(y))) \wedge f(g(y),y) = y \wedge x = g(y)$

Answer: the subterms that show up are

$$\{\{x\},\{y\},\{g(y)\},\{f(x,y)\},\{f(y,x)\},\{f(y,g(y))\},\{f(g(y),y)\},\{g(f(y,g(y)))\}\} \tag{3}$$

the equalites asserted makes

$$\{\{x,g(y)\},\{y,f(g(y),y)\},\{f(x,y),f(y,x)\},\{f(y,g(y))\},\{g(f(y,g(y)))\}\} \tag{4}$$

note $x \sim g(y)$ and $y \sim y$.
thus

$$\{\{x,g(y)\},\{y,f(g(y),y),f(x,y),f(y,x)\},\{f(y,g(y))\},\{g(f(y,g(y)))\}\} \tag{5}$$

also $y \sim y$ and $x \sim g(y)$.
which gives

$$\{\{x,g(y)\},\{y,f(g(y),y),f(x,y),f(y,x),f(y,g(y))\},\{g(f(y,g(y)))\}\} \tag{6}$$

finally since $y \sim f(y,g(y))$

$$\{\{x,g(y),g(f(y,g(y)))\},\{y,f(g(y),y),f(x,y),f(y,x),f(y,g(y))\}\} \tag{7}$$

to check inequalities
$g(y)$ is in class $\{x,g(y),g(f(y,g(y)))\}$ but $g(f(y,g(y)))$ is in the same class, therefore there is a contradiction.

2. (70 pts.) Lazy SMT solving

Consider the following Linear Integer Arithmetic (LIA) formula ϕ_{SMT}:

$$\begin{aligned}&(x > 5 \ \vee \ \neg(y > 3))\\ \wedge&(x \leq 4 \ \vee \ x + y < 9)\\ \wedge&(y > 3 \ \vee \ y > x)\\ \wedge&(x > 3)\end{aligned}$$

We'll use the lazy SMT algorithm to solve this. You can play the part of the LIA conjunction decision procedure yourself (e.g. you can tell that $x > 10 \wedge x < 3$ is UNSAT, since no integer is simultaneously greater than 10 and less than 3). You don't need to show the internals of the SAT solving: feel free to use any SAT solver, e.g. the online version of CryptoMiniSat here.

(a) What is the Boolean abstraction ϕ_{SAT} of ϕ_{SMT}?

Answer:

let

- $A := (x > 5)$
- $B := (y > 3)$
- $C := (x \leq 4)$
- $D := (x + y < 9)$
- $E := (y > x)$
- $F := (x > 3)$

we get

$$\phi_{\text{SAT}} = (A \vee \neg B) \wedge (C \vee D) \wedge (B \vee E) \wedge F \tag{8}$$

(b) Give a satisfying assignment of ϕ_{SAT}, and the corresponding conjunction of LIA literals.

Answer:

let

$$A = \top, \quad B = \top, \quad C = \top, \quad D = \bot, \quad E = \bot, \quad F = \top \tag{9}$$

which is the same as

$$(x > 5) \wedge (y > 3) \wedge (x \leq 4) \wedge (x + y \geq 9) \wedge (y \leq x) \wedge (x > 3) \tag{10}$$

(c) Is this conjunction satisfiable? If so, give a satisfying assignment. If not, give a theory lemma which has as few literals as possible.

Answer: no it is not. x cant be greater than 5 and less than 4.

new theory lemma:

$$\neg(x > 5) \vee \neg(x \leq 4) \tag{11}$$

(d) Continue the lazy SMT algorithm until you find a satisfying assignment for ϕ_{SMT} or prove it unsatisfiable. Show each solution you find to the Boolean abstraction and each theory lemma you add.

Answer:

$$A = \top, \quad B = \top, \quad C = \bot, \quad D = \top, \quad E = \bot, \quad F = \top \tag{12}$$

which gives

$$(x > 5) \wedge (y > 3) \wedge (x > 4) \wedge (x + y < 9) \wedge (y \leq x) \wedge (x > 3) \tag{13}$$

but this also breaks because of

$$\neg(x > 5) \vee \neg(y > 3) \vee \neg(x + y < 9) \tag{14}$$

$$A = \bot, \quad B = \bot, \quad C = \bot, \quad D = \top, \quad E = \top, \quad F = \top \tag{15}$$

$$(x \leq 5) \wedge (y \leq 3) \wedge (x > 4) \wedge (x + y < 9) \wedge (y > x) \wedge (x > 3) \tag{16}$$

which breaks

$$(x \leq 4) \vee (y > 3) \vee \neg(y > x) \quad (\text{or } C \vee B \vee \neg E) \tag{17}$$

$$A = \bot, \quad B = \bot, \quad C = \top, \quad D = \top, \quad E = \top, \quad F = \top \tag{18}$$

$$(x \leq 5) \wedge (y \leq 3) \wedge (x \leq 4) \wedge (x + y < 9) \wedge (y > x) \wedge (x > 3) \tag{19}$$

which adds

$$\neg(x \leq 4) \vee \neg(y \leq 3) \vee \neg(y > x) \tag{20}$$

but the final formula with all lemmas is unsat so the original expression is unsat.